



Apple MdZAT5 mediates root development under drought stress

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ABSTRACT

Root plays an important role in plant drought tolerance, especially in horticultural crops like apples. However, the crucial regulator and molecular mechanism in root development of apple trees under drought are not well unknown. Cys2/His2-type Zinc-finger proteins are essential for plant response to drought, while the members of C2H2 Zinc-finger proteins in apple are largely unknown. In this study, we identified the members of the C1-2i subclass family of C2H2 Zinc-finger proteins in apple (*Malus* × *domestica*). Among them, MdZAT5 is significantly induced in apple roots under drought conditions and positively regulates apple root development under drought. Further investigation revealed that MdZAT5 positively regulates root development and root hydraulic conductivity by mediating the transcription level of *MdMYB88* under drought stress. Taken together, our results demonstrate the importance of MdZAT5 in root development under drought in apple trees. This finding provides a new candidate direction for apple breeding for drought resistance.

1. Introduction

As sessile organisms, plants frequently endure damage from abiotic environmental factors, with drought being the primary environmental factor impeding vegetative growth and development (González, 2023), especially horticultural crops. The quality and yield of temperate fruits have recently suffered significant losses due to frequent drought events (Zhang et al., 2023). These are of great concern to commercial orchards, as drought stress can hinder the growth of fruit trees and reduce fruit quality and yield (Devin et al., 2023). Therefore, it is necessary to explore increasing drought responsive genes to enhance the drought tolerance of horticultural crops and develop new, drought-tolerant varieties.

The complexity of plant drought response involves the integration of multiple response and adaptation mechanisms (Li et al., 2023). Under mild water shortage stress, plants maintain normal growth by enhancing

root activity and architecture, thereby increasing water absorption capacity and reducing water loss through stomatal closure (Seki et al., 2007). In addition to morphological and physiological responses, plants alter gene expression at the molecular level. Transcription factors, essential regulatory proteins, mediate the transcription of key structural genes in plant drought response (Singh et al., 2002). Various transcription factor families, such as myeloblastosis (MYB) related, dehydration responsive element binding (DREB), Zinc-finger proteins etc. play crucial roles in plant drought response (Manna et al., 2021; Han et al., 2020).

Cys2/His2-type Zinc-finger proteins serve as key transcriptional regulators in plant responses to various stress conditions, including extreme temperature, drought, oxidative stress, etc. (Han et al., 2020; Li et al., 2022; Liu et al., 2022). In the *Arabidopsis* genome, 176 members belong to the zinc-finger proteins (Englbrecht et al., 2004), many of which are involved in plant growth and development (Wang et al.,

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2019). C2H2-type zinc finger TFs are classified into three sets (A, B, and C), with each set having its own subsets (Ciftci-Yilmaz and Mittler, 2008). Among them, the C1-2i subclass family, which contains 20 members in *Arabidopsis*, is notable as most of its members participate in plant biotic and abiotic stress responses (Davletova et al., 2005; Ciftci-Yilmaz and Mittler, 2008; Kodaira et al., 2011; Tang and Luo, 2018; Yang et al., 2021). To date, the C1-2i subclass family in apple has not yet been identified.

Both above and belowground plant components can respond to drought, with plants exhibiting complex above- and below-ground trait adjustments to sustained drought conditions (Rowland et al., 2023). A recent study demonstrated that MdMYB88/124 functioned in both above and belowground apple components under drought conditions (Geng et al., 2018; Xie et al., 2021). These proteins bind to the promoters of *MdVND6* and *MdMYB46*, regulating xylem development and secondary cell wall formation, thereby regulating apple root drought resistance (Geng et al., 2018). During short-term drought stress, MdMYB88 and MdMYB124 enhanced apple drought tolerance by promoting ABA biosynthesis through interaction with the *MdNCED3* gene promoter (Xie et al., 2021). Previously, we conducted an RNA-seq analysis on *M. sieversii* roots subjected to PEG treatment (Geng et al., 2019), leading to the identification of MdZAT5, a C2H2 transcription factor (a member of C1-2i subclass family) induced by drought in apple leaves (Bao et al., 2022). MdZAT5 improves apple drought tolerance by regulating drought-responsive mRNAs and miRNAs (Bao et al., 2022). However, the role of ZAT5 in plant root drought response remains uncertain.

In this study, we identified and characterized the C1-2i subclass family members in apple, examining their properties (length, chromosomal distribution, and gene structure). Moreover, we examined the protein characteristics of this family, including length, conserved domains and motifs, and phylogenetic relationships with *Arabidopsis* and rice. Furthermore, we demonstrated that MdZAT5, which is significantly induced by drought, functions as a positive regulator of apple root development under drought stress. Additionally, we found that MdZAT5 positively regulates root development under drought stress by regulating the *MdMYB88* transcription level.

2. Materials and methods

2.1. Plant materials, growth conditions, and stress treatment

The MdZAT5 RNAi plants were generated previously (Bao et al., 2022). For long-term drought treatment, 20 each of tissue-cultured GL-3 seedlings and MdZAT5 transgenic plants were rooted for 6 weeks under rooting medium (2.215 g/L MS salts, 7.5 g/L agar, 0.5 mg/L IBA, 0.5 mg/L IAA, and 20 g/L sucrose, pH = 5.8–6.0) and transferred to pots (30 cm × 18 cm). After two months, these plants were placed in a naturally lit greenhouse maintained at 22–37 °C and 37–55% relative humidity. After another month, seedlings were divided into well-watered (n = 10) and long-term drought treatment groups (n = 10) for both GL-3 and MdZAT5 transgenic plants. The well-watered group plants received daily irrigation to maintain field capacity of 75–85%, while the drought group plants were maintained at 45–55%. After 2-month drought treatment, shoots and roots were harvested for morphology analysis. Morphological analysis involved direct measurement of plant height, stem diameter, and dry weights of roots and shoots. Three independent experiments were performed (each experiment performed ten biological replicates), and the representative replicate was shown in Figure. Each plant is a biological replicate.

For PEG treatment, tissue cultured GL-3 or *M. sieversii* seedlings were rooted for 6 weeks. Then these plants were cultivated hydroponically for one additional month and treated with 20% PEG 8000 for 6 h. Plant leaves and roots were collected and five leaves or roots were used as a biological replication. Three biological replicates were applied for each of the control and the treatment. Total RNA was extracted and used for

RNA-seq and qRT-PCR. Additional Tissue cultured GL-3 seedlings were rooted and transferred to pot. After well-watered, GL-3 roots were collected under drought treatment for 0 or 7d.

2.2. Detection of leaves stable carbon isotope composition ($\delta^{13}C$)

For the detection of leaves $\delta^{13}C$, mature leaves were collected and subjected to oven-drying at 105 °C for 30 min, followed by 65 °C for five days. Subsequently, the dried leaves were ground, filtered through an 80 holes/cm² sieve, and $\delta^{13}C$ was determined with an elementary analysis-isotope ratio mass spectrometer as described by Wang et al. (2018).

2.3. Measurement of root or shoot hydraulic conductivity

Root or shoot hydraulic conductivity was measured by high pressure flow matter (HPFM, Dynamax, Houston) as described by Geng et al. (2018). Following a 2-month drought treatment, GL-3 and MdZAT5 RNAi plants were bisected 2–3 cm above ground, and the sections were then connected to the HPFM. Root hydraulic conductivity was assessed using the transient method, whereas shoot hydraulic conductivity was evaluated using the quasi steady-state method.

2.4. qRT-PCR

The RNA was extracted by CTAB method which was previously described (Xie et al., 2018). qRT-PCR was performed as described by Niu et al. (2019). The primers used for this study were listed in Supplemental Table 4. The relative expression calculated method is $2^{-\Delta\Delta CT}$ (Chang et al., 1993). MdMDH was used as reference gene.

2.5. ChIP-qPCR

ChIP-qPCR assay was performed as described by Xie et al. (2018). The Leaves from four weeks old GL-3 tissue culture seedling were cross-linked in 1% (w/v) formaldehyde. Then chromatin were isolated and sonicated. Immunoprecipitation was performed with an anti-MdZAT5 antibody previously generated (Bao et al., 2022) overnight at 4 °C. After precipitating using ChIP-grade protein A/G agarose beads (Thermo Scientific™, USA), washing using different washing buffers four times, reversing crosslink, and protein removal, the recovered DNA was used for qPCR. Primers used for ChIP-qPCR are listed in Supplemental Table 4. MdMDH was used as reference gene.

2.6. Bioinformatic analysis

Based on the previous identification of C1-2i gene family members in *Arabidopsis thaliana*, blastp in TBtools-II v2.003 were used to identify candidate C1-2i genes in the latest Apple genome (https://iris.angers.inra.fr/gddh13/downloads/GDDH13_11_formatted.fasta.bz2) (Chen et al., 2023). The C1-2i protein sequences from *Arabidopsis* were download from TAIR (<https://www.arabidopsis.org>). Then based on the C1-2i gene family members in Apple which were identified from *Arabidopsis thaliana*, blastp in TBtools-II v2.086 were used to identify candidate C1-2i genes in the latest Rice genome (https://ftp.ebi.ac.uk/ensemblgenomes/pub/release-58/plants/fasta/oryza_sativa/dna/Oryza_sativa_IRGSP-1.0.dna.toplevel.fa.gz). The alignments of *Arabidopsis* C1-2i, Apple C1-2i and Rice C1-2i were used muscle software. The alignments file was then submitted to the following phylogenetic tree construction using the Maximum Likelihood (ML) tree method and Jones-Taylor-Thornton(JTT) + Gamma distribution(G) model with 1000 bootstrap replications in MEGA 11 (Tamura et al., 2021). The phylogenetic tree was annotated and depicted using FigTree v1.4.4 (<http://tree.bio.ed.ac.uk/software/figtree/>) and chiplot (<https://www.chipplot.online>). CDD search was used to verify the conserved domains. Mw and pI were evaluated using “Compute I/Mw” online tools (https://web.expasy.org/compute_pi/). MEME online tool (<https://mem>

e-suite.org/meme/) was used to predicted the motif. The duplication events in C1-2i family were determined using MCScanX (Multiple Collinearity Scan toolkit) program in TBtools. Gene structure and motif of Apple C1-2i were also visualized using TBtools.

2.7. EMSA

MdZAT5-GST protein was previously generated (Bao et al., 2022). The EMSA assay was performed according to the manual of LightShift Chemiluminescent EMSA Kit (#89880, Thermo Scientific™, USA). The oligonucleotide probes labeled with biotin are listed in Supplemental Table 4.

2.8. Yeast-one hybrid analysis

The Fragment a and b of MdMYB88 promoter were individually cloned into pAbAi vector, then linearized and transformed into yeast strain Y1H Gold, followed by growth selection on SD medium without uracil (Clontech, USA) for 3–5 d under 28 °C. The Aureobasidin A (Aba) concentration was determined using positive clones, and MdZAT5-pGADT7 vector then was transformed into Y1H Gold competent cells (*proMdMYB88* a-pAbAi or *proMdMYB88* b-pAbAi). Yeast cells were grown on SD-Leu/AbA. pGADT7 empty vector served as a negative control. Primers used were listed in Supplemental Table 4.

3. Results

3.1. Identification of C1-2i gene family members

Sequence blast and analysis led to the identification of 43 C1-2i family genes, which were subsequently named MdZAT1–MdZAT43. Essential gene information is presented in Supplemental Table 1. The protein sequence length varies from 179 (MdZAT12) to 599 (MdZAT9) amino acids. The molecular weight ranges from 18,704.75 Da

(MdZAT12) to 65,535.66 Da (MdZAT9), with the theoretical isoelectric point of these proteins falling between 5.42 and 10.7.

To study the evolutionary relationship between the *MdZATs* genes of apple, rice, and *Arabidopsis*, a phylogenetic tree was constructed (Fig. 1). Based on the results of Figs. 1 and 2, 43 MdZATs were divided into eight groups (1–8) using the motif appears regularly of each member and the distance on the evolutionary tree of then. Among them, group 6 had seven members, while groups 1, 2, and 7 had six each. Groups 5 and 8 each contained five MdZATs, and groups 3 and 4 each contained four.

3.2. Structure and motif analysis of MdZATs gene in apple

The gene structure analysis of the *MdZAT* genes is displayed in Supplemental Fig. 1, revealing that none of the apple *MdZAT* genes possessed introns. The protein conserved motif analysis results for apple MdZATs are presented in Fig. 2. As shown, we identified 10 conserved motifs (Supplemental Table 2). Among them, the group 5 members all contain motifs 1, 2, 4, 5, and 6. Group 6 members all have motifs 1, 2, 4, and 5. Some group 8 members have only motif 1 or motifs 1 and 10, thereby suggesting that motifs 1 and 10 represent the evolutionary beginning of *MdZAT* genes in apple.

3.3. Chromosome localization and collinearity analysis

Supplemental Fig. 2 presents the chromosome localization results. Uneven distribution of the 43 MdZAT genes across apple chromosomes was observed, with chromosomes 3 and 7 hosting the highest gene count (five each). Chromosomes 7, 11, and 13 exhibited coaggregation, potentially resulting from chromosome tandem duplication. Chromosomes 1, 11, and 13 each harbored four genes; chromosomes 2, 5, 6, and 10 contained two genes; and chromosomes 8, 9, 14, and 15 had one gene each. Chromosomes 4 and 12 contained three genes. However, one gene's chromosomal location remained unpredicted due to genome assembly complexities. Fig. 3 displays the collinearity results, indicating

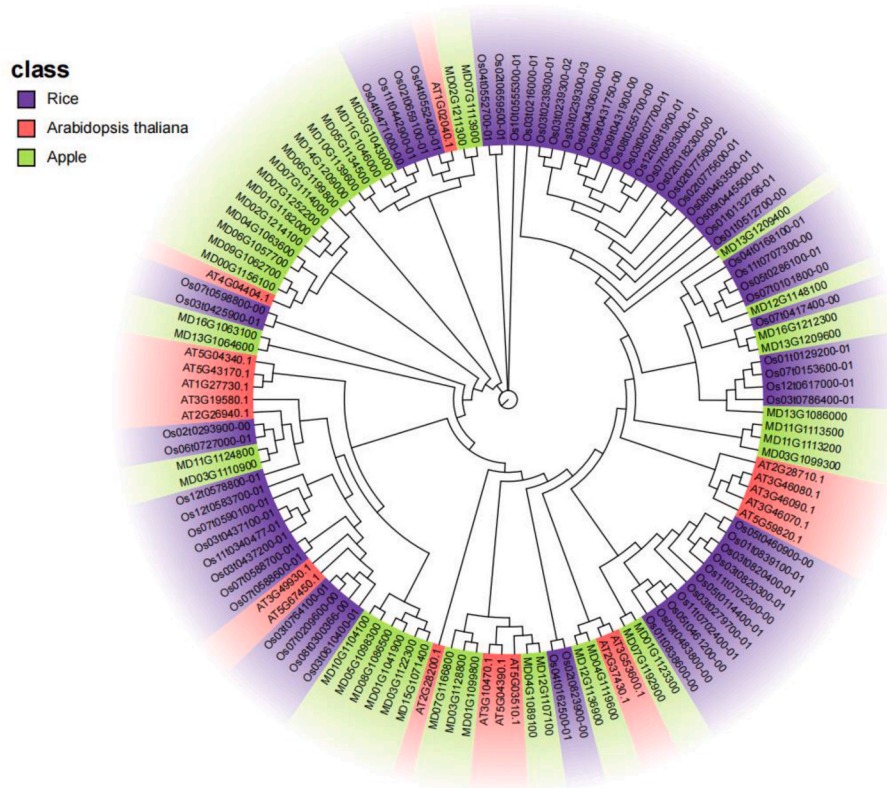


Fig. 1. Phylogenetic tree of C1-2i subclass family in *Arabidopsis thaliana*, Rice, and Apple.

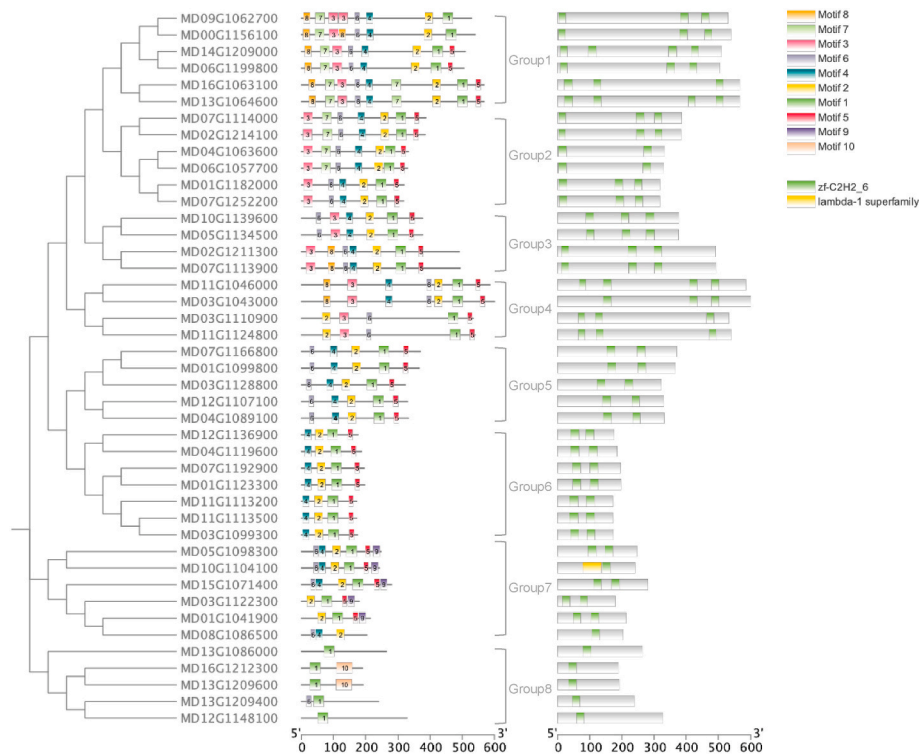


Fig. 2. Motif analysis of C1-2i subclass family members in apple. The motif types are represented by different colored squares.

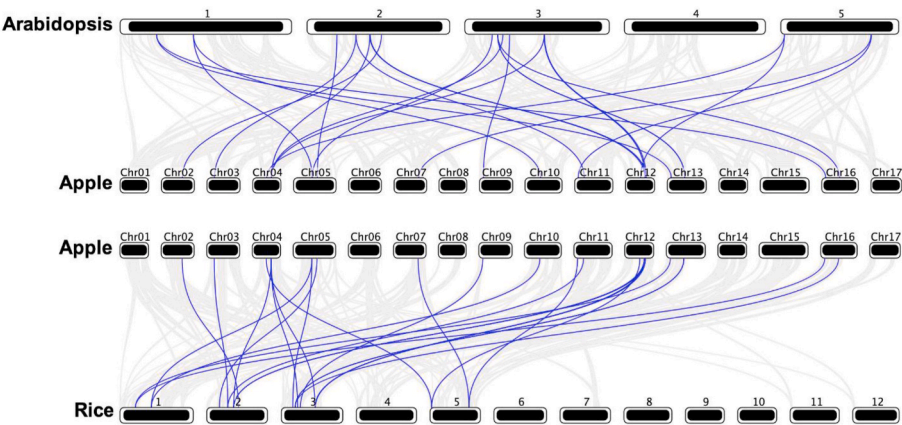


Fig. 3. Collinearity analysis of C1-2i members in apple, *A. thaliana*, and rice. The grey lines in the background illustrate the collinearity of the entire genome, while the blue lines highlight the collinearity of the C1-2i family genes.

20 pairs of syntenic ZAT genes shared between apple and *Arabidopsis*, and 21 pairs between apple and rice.

3.4. Expression pattern of MdZAT genes in apple roots under drought stress

The genes' response to drought suggests their involvement in drought stress. Previous RNA-seq data presented in Supplemental Table 3 revealed the read counts of MdZAT genes in GL-3 plants (Bao et al., 2022). Notably, MD08G1086500, MD03G1128800, MD01G1041900, MD15G1071400, and MD13G1064600 exhibited significantly higher read counts than the others. Subsequent qRT-PCR analysis detected the response of these candidate genes under PEG treatment, revealing that MD03G1128800 (*MdZAT5*) was significantly upregulated in apple roots (Fig. 4). This finding suggests a potential key role for MdZAT5 in roots under drought stress.

3.5. MdZAT5 positively regulates drought tolerance under long-term drought treatment

To identify the expression pattern of MdZAT5 under drought, we detected its expression in *M. sieversii* roots under PEG treatment and in GL-3 roots under drought for 0 or 7 d. The results further indicated that MdZAT5 is drought-inducible in apple roots (Supplemental Figs. 3a and b). To further examine the function of MdZAT5 in roots under drought conditions, we analyzed apple root phenotypes of non-transgenic plants GL-3 and previously generated MdZAT5 RNAi plants (Bao et al., 2022). We subjected seedlings to long-term drought treatment, maintaining a field capacity of 45–55% for 60 d, and measured plant height and stem diameter. As shown in Fig. 5, the 60 d drought treatment significantly affected plant height; MdZAT5 RNAi plants were shorter than GL-3 under long-term drought conditions, although their heights and stem diameters were similar in control conditions (Fig. 5a–c). After 60 d of

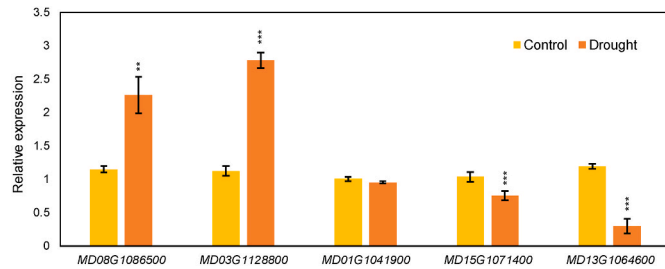


Fig. 4. Expression level of *MD08G1086500*, *MD03G1128800*, *MD01G1041900*, *MD15G1071400*, and *MD13G1064600* in GL-3 roots under PEG treatment. Two-month-old GL-3 plants were cultivated hydroponically for one additional month and treated with 20% PEG 8000 for 6 h. Error bars indicate standard deviation (n = 3). Student's two tailed t-test was performed and statistically significant differences were indicated by * (P < 0.05), ** (P < 0.01) and *** (P < 0.001).

long-term drought treatment, we measured the biomass accumulation of these plants. The biomass accumulation of *MdZAT5* RNAi plants decreased by 42–50%, while GL-3 plants showed only a 25–26%

reduction (Fig. 6a). Water use efficiency (WUE), an indicator of plant drought tolerance, is physiologically and genetically linked to stable carbon isotope composition ($\delta^{13}\text{C}$) in plant leaves (Farquhar et al., 1982; Chen et al., 2011; Medrano et al., 2016). *MdZAT5* RNAi plants had lower $\delta^{13}\text{C}$ than GL-3 plants (Fig. 6b). These results suggested that *MdZAT5* positively regulates apple's drought response under long-term drought treatment.

3.6. *MdZAT5* positively regulates root morphology and root hydraulic conductivity of apple under long-term drought treatment

To investigate *MdZAT5*'s role in root development under drought conditions, we measured the root morphology and quantitated morphological traits following a 2-month drought treatment. Fig. 7 shows that drought-treated *MdZAT5* RNAi plants had smaller roots than GL-3 plants (Fig. 7a). Both root and shoot dry weights were significantly lower in *MdZAT5* RNAi plants than in GL-3 plants (Fig. 7b and c), although the root-to-shoot ratio remained similar (Supplemental Fig. 4b). These findings suggest that *MdZAT5* enhances drought tolerance, partly by regulating root development.

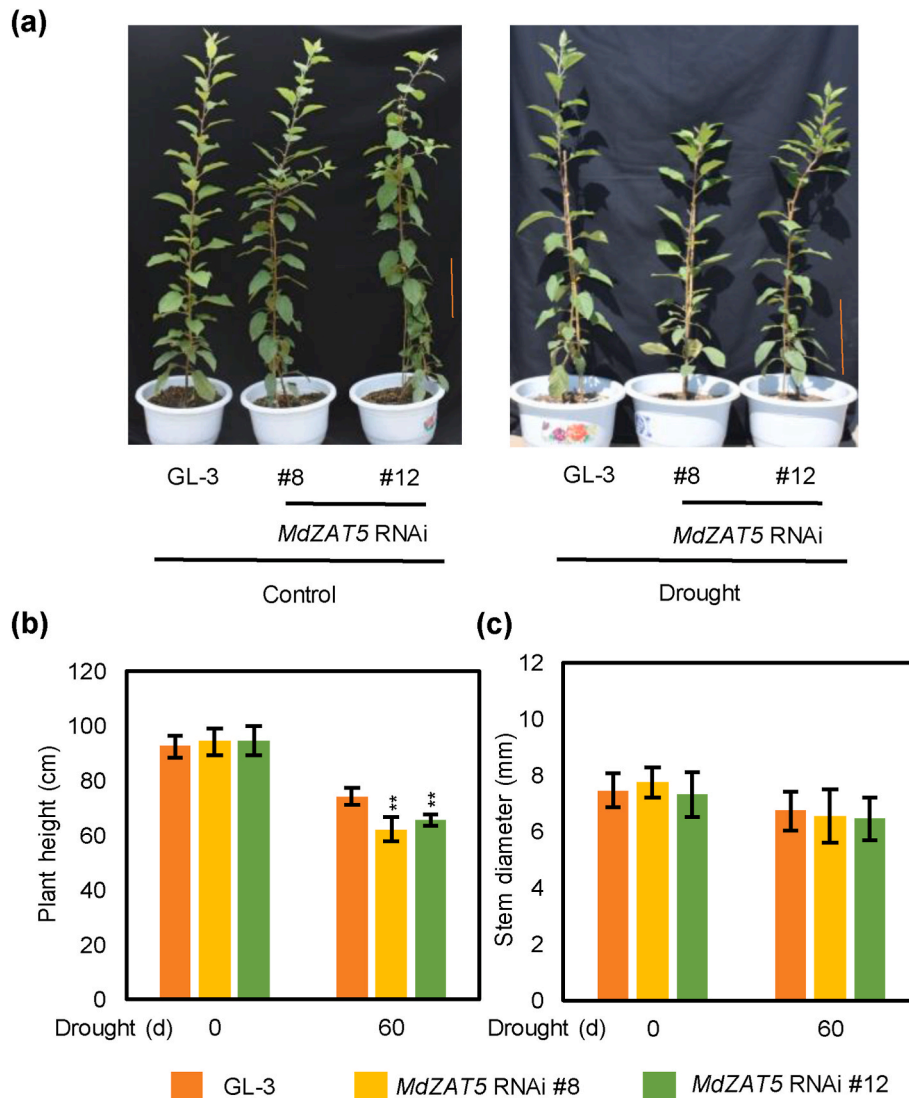


Fig. 5. Morphology of *MdZAT5* transgenic plants under long-term drought conditions. (a) Morphology of nontransgenic plants (GL-3) and *MdZAT5* RNAi plants under long-term drought treatment. Plants were subjected to long-term drought stress for 60 days in a greenhouse. (b) Plant height (c) stem diameter of GL-3 and *MdZAT5* RNAi plants under long-term drought treatment. Bars = 15 cm in (a). Error bars indicate standard deviation (n = 10 in (b) and (c)). Student's two tailed t-test was performed and statistically significant differences were indicated by ** (P < 0.01).

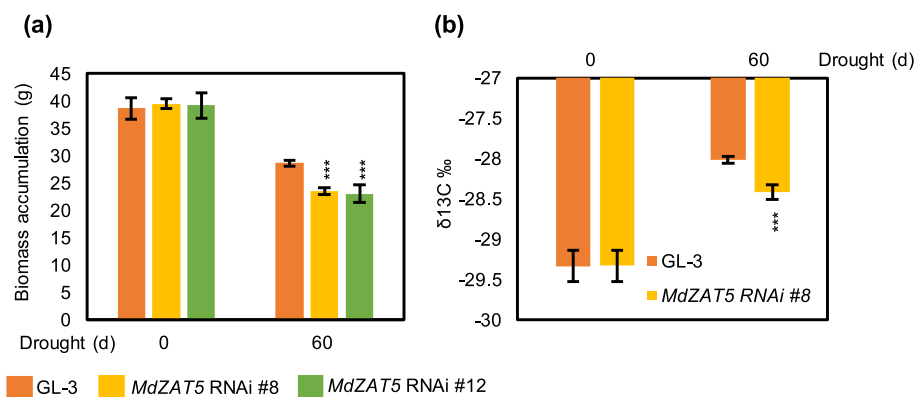


Fig. 6. Biomass accumulation and stable carbon isotope composition ($\delta^{13}\text{C}$) of GL-3 and *MdZAT5* RNAi plants under long-term drought treatment. Plants were subjected to long-term drought stress for 60 days in a greenhouse. Error bars indicate standard deviation ($n = 10$). Student's two tailed t -test was performed and statistically significant differences were indicated by *** ($P < 0.001$).

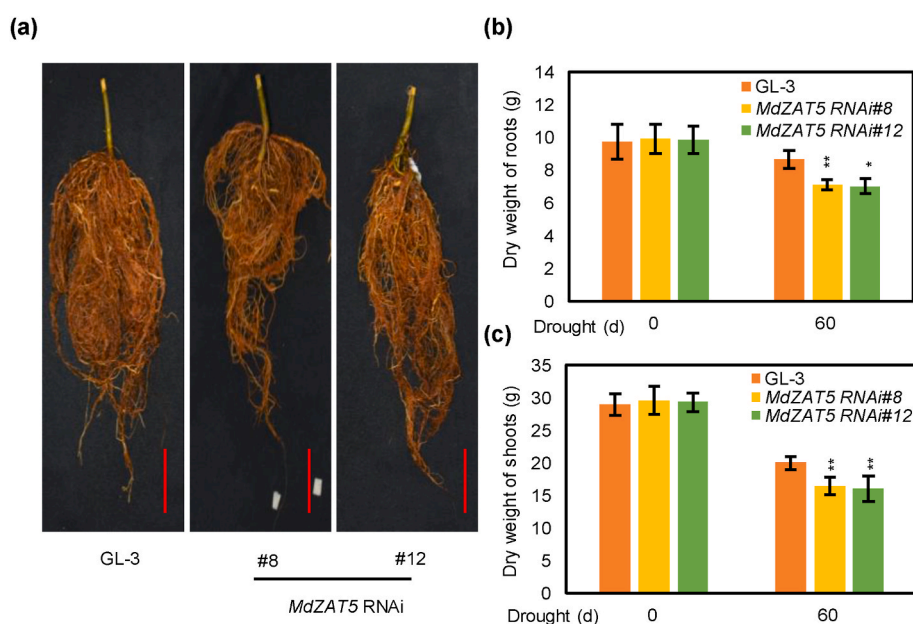


Fig. 7. Root morphology and quantitation of morphological traits of *MdZAT5* transgenic plants under long-term drought conditions. (a) Root morphology of GL-3 and transgenic *MdZAT5* RNAi plants under long-term drought conditions. (b) Dry weight of roots. (c) Dry weight of shoots. Plants were subjected to long-term drought stress for 60 days in a greenhouse. Bars = 5 cm in (a). Error bars indicate standard deviation ($n = 10$). Student's two tailed t -test was performed and statistically significant differences were indicated by ** ($P < 0.01$) and * ($P < 0.05$).

Roots primarily transport water and mineral elements to above-ground parts (Lux and Rost, 2012). Hydraulic conductivity, an indicator of plants' water transport ability, decreases in roots and shoots under drought stress (Geng et al., 2018). The significant difference of root morphology of drought-stressed GL-3 and *MdZAT5* RNAi transgenic plants led us to examine root hydraulic conductivity. Under drought stress, the *MdZAT5* RNAi plant roots displayed significantly lower hydraulic conductivity than GL-3 (Fig. 8). Additionally, *MdZAT5* RNAi plants had significantly lower shoot hydraulic conductivity compared to GL-3 under drought conditions (Supplemental Fig. 5). These results indicate that *MdZAT5* is vital for maintaining apple root morphology and hydraulic conductivity during prolonged drought treatment.

3.7. *MdZAT5* positively regulates the transcription of *MdMYB88* under drought stress

MdZAT5 can bind to specific cis-elements in the promoters of target genes. To investigate its role in root development under drought conditions, we analyzed the DAP-seq data for *MdZAT5* (Bao et al., 2022)

and identified *MdMYB88*, a positive regulator of root development, as a potential target. Comparison of *MdMYB88* transcript levels in the roots of GL-3 and *MdZAT5* transgenic plants revealed comparatively lower expression in *MdZAT5* RNAi plants (Fig. 9a), thereby suggesting that *MdZAT5* activates *MdMYB88* expression under drought conditions. Analysis of the *MdMYB88* promoter sequence revealed three *MdZAT5* recognition motifs (one each of ACATA, TCACAG, and TCACTG; Supplemental Fig. 6). The Yeast-one hybrid analysis showed that *MdZAT5* binds to the fragment a and b of *MdMYB88* promoter (Fig. 9c). The ChIP-qPCR assay using *MdZAT5* native antibody confirmed that *MdZAT5* binds directly to the fragments in the *MdMYB88* promoter (Fig. 9d). EMSA suggested that *MdZAT5* could directly bind to the *MdMYB88* promoter region (Fig. 9e). Thus, these findings suggest that *MdZAT5* regulates root development under drought conditions by directly activating *MdMYB88* transcription.

4. Discussion

C2H2 zinc finger proteins are master regulators of plant development

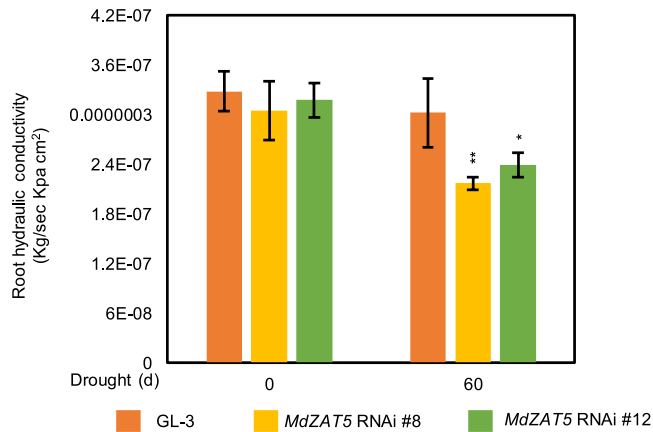


Fig. 8. Root hydraulic conductivity of GL-3 and *MdZAT5* RNAi plants under long-term drought conditions. Plants were subjected to long-term drought stress for 60 days in a greenhouse. Error bars indicate standard deviation ($n = 10$). Student's two tailed t -test was performed and statistically significant differences were indicated by ** ($P < 0.01$) and * ($P < 0.05$).

and abiotic stress response (Kiebowicz-Matuk, 2012; Han et al., 2021). In our study, we identified 43 members of the C1-2i subclass family in apples, compared to the 20 members in the *Arabidopsis thaliana* C1-2i

subclass family (Englbrecht et al., 2004). These apple members can be divided into eight groups (1–8), with varying distributions of gene numbers. Group 8, containing fewer motifs or domains, may represent an earlier evolutionary stage. All *MdZAT* genes lack introns in their CDS region and have very short sequences, typically <1000 bp, as shown in Supplemental Fig. 2. These gene structures facilitate rapid transcription in specific tissues or conditions, suggesting that *MdZAT* genes play crucial roles in plant development. Drought significantly limits horticultural plant development, particularly in apple species. To investigate the apple's candidate drought-responsive genes, we examined the expression patterns of apple *MdZAT* genes under simulated drought conditions. Upon analyzing the reads counts of these *MdZAT* genes, we found no reads counts for several genes in GL-3 plants, including MD13G1209600, MD02G1211300, MD12G1148100, MD05G1134500, MD11G1124800, MD12G1136900, MD16G1212300, MD03G1110900, MD01G1182000, MD10G1139600, MD13G1209400, and MD07G1252200 (Supplemental Table 3). Subsequently, we focused on the expression pattern of the first four genes with higher reads counts, namely MD08G1086500, MD03G1128800, MD01G1041900, MD15G1071400, and MD13G1064600, in GL-3 roots under PEG-simulated drought treatment. The results demonstrated that *MdZAT5* exhibited the most significant upregulation under PEG treatment in apple roots (Fig. 4, Supplemental Fig. 3), suggesting a potential role for *MdZAT5* in drought response.

In our previous study, although we characterized the function of *MdZAT5* in drought stress in aboveground apple tissues (Bao et al.,

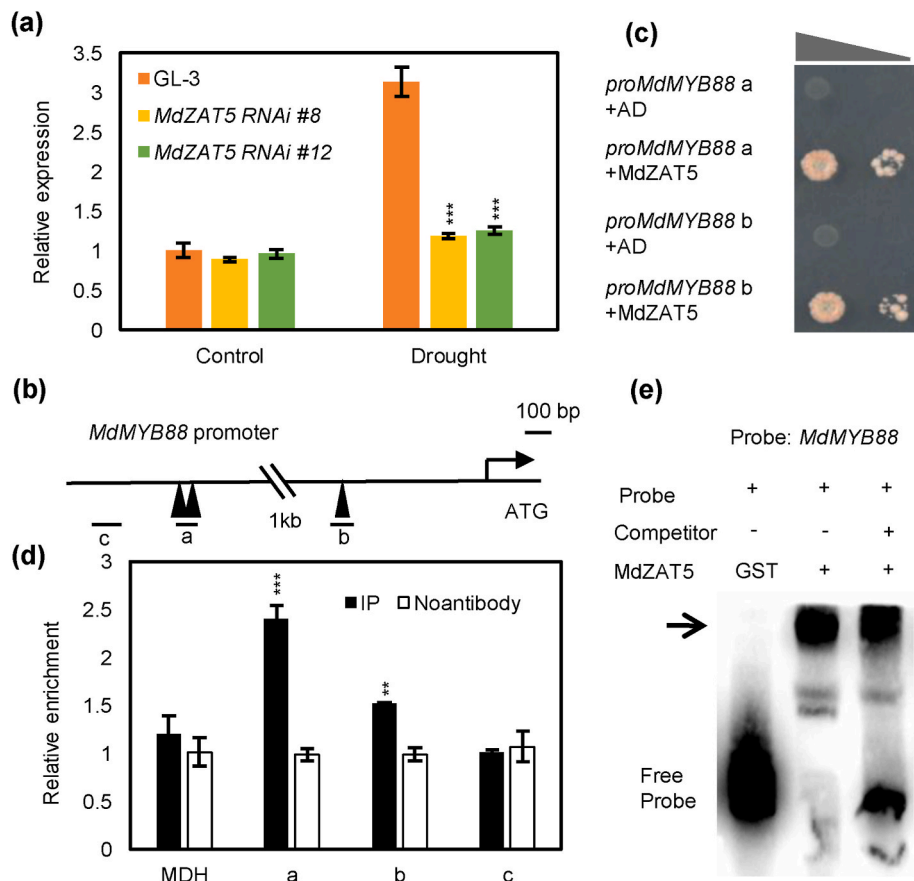


Fig. 9. *MdZAT5* regulates root development under drought stress by binding to the promoter region of *MdMYB88*. (a) Relative mRNA level of *MdMYB88* in roots of GL-3 and *MdZAT5* transgenic plants under control or drought treatment for 7d. (b) Diagram of *MdMYB88* promoter. Fragment a contains one TCACAG and one AACTA motif; Fragment b contains one TCACTG motif; Fragment c serves as a negative control. (c) Yeast-one hybrid analysis of interaction between *MdZAT5* and *MdMYB88* promoter. Yeast cells were grown on SD-Leu with 300 ng/mL AbA. (d) ChIP-qPCR analysis of *MdZAT5* bound to the promoter of *MdMYB88*. Fragment c, MDH, and no antibody serve as negative controls. (e) EMSA analysis of *MdZAT5* binding to *MdMYB88* promoter. Arrowheads indicate protein-DNA complex. Error bars indicate standard deviation ($n = 3$). Student's two tailed t -test was performed and statistically significant differences were indicated by ** ($P < 0.01$) and *** ($P < 0.001$).

2022), its role in underground drought response remains unexplored. In this study, we observed *MdZAT5*'s drought inducibility in apple roots (Fig. 4, Supplemental Fig. 3), prompting an investigation into its function under drought conditions by examining root phenotypes. Consequently, a 2-month long-term drought treatment was conducted. Our findings revealed that *MdZAT5* silencing significantly diminished apple drought tolerance. We then assessed the physiological indices of GL-3 and *MdZAT5* RNAi plants. Under drought conditions, the plant height, biomass accumulation, $\delta^{13}\text{C}$, dry weight of roots and shoots, and root and shoot hydraulic conductivity of *MdZAT5*-RNAi plants were significantly lower than those of GL-3 plants (Figs. 5–8 and Supplemental Fig. 5). Although no significant difference was observed in the root-to-shoot ratio between *MdZAT5* RNAi plants and GL-3 under drought stress, this could be attributed to the similar dry weight differences in shoots and roots between *MdZAT5* RNAi plants and GL-3 plants under drought. These findings suggest that *MdZAT5* mediates drought tolerance by regulating root architecture and root hydraulic conductivity.

The transcription factors bind specific motifs in target gene promoters, playing a crucial role in root development under drought conditions (Lee et al., 2016; Zhao et al., 2019; Wan et al., 2023). For instance, MdMYB88 binds to the promoters of *MdVND6* and *MdMYB46*, activating their transcription to enhance xylem development and secondary cell wall formation in roots under drought stress (Geng et al., 2018). Long-term drought treatment reduces dry weight and hydraulic conductivity of roots and shoots in *MdMYB88/124* RNAi plants compared to GL-3 (Geng et al., 2018). In this study, we identified MdMYB88 from previously analyzed DAP-seq data of *MdZAT5*. Moreover, we found three *MdZAT5* binding sites in the *MdMYB88* promoter region (Supplemental Fig. 6). The qRT-PCR, Yeast-one hybrid analysis, EMSA and ChIP-qPCR results indicate that *MdZAT5* binds to the *MdMYB88* promoter, thereby activating its transcript under drought conditions (Fig. 9). Under control conditions, *MdMYB88* expression does not significantly differ between GL-3 and *MdZAT5* RNAi plants, thus suggesting a reason for the similar root architecture, dry weight, and hydraulic conductivity in *MdZAT5* RNAi plants and GL-3.

5. Conclusions

Our results identified a total of 43 putative C2H2 C1-2i family genes within *Malus domestica* genome. The gene structures, conservative domains, and conservative motifs of these members were analyzed and determined. Then the evolutionary relationships between these C2H2 C1-2i family members in *Malus domestica*, *Arabidopsis* and rice have been examined. Among these genes, *MdZAT5* is most upregulated by PEG-simulated drought in apple roots. Further, we demonstrated that *MdZAT5* modulates root development by activating *MdMYB88* under drought conditions, thus regulates drought response.

CRedit authorship contribution statement

Yutian Zhang: Data curation. **Jieqiang He:** Formal analysis. **Gege Qin:** Data curation. **Kecheng Yang:** Data curation. **Pengxiang Chen:** Data curation. **Chundong Niu:** Formal analysis. **Xuwei Li:** Formal analysis. **Chuang Mei:** Investigation. **Jiangbo Wang:** Writing – original draft, Project administration. **Qingmei Guan:** Project administration. **Chana Bao:** Writing – review & editing, Writing – original draft, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my data at the Attach File step.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.plaphy.2024.108833>.

References

- Bao, C., Qin, G., Cao, F., He, J., Shen, X., Chen, P., Niu, C., Zhang, D., Ren, T., Zhi, F., Ma, L., Ma, F., Guan, Q., 2022. *MdZAT5* regulates drought tolerance via mediating accumulation of drought-responsive miRNAs and mRNAs in apple. *New Phytol.* 236 (6), 2131–2150. <https://doi.org/10.1111/nph.18512>.
- Chang, S., Puryear, J., Cairney, J., 1993. A simple and efficient method for isolating RNA from pine trees. *Plant Mol. Biol. Rep.* 11, 113–116.
- Chen, C., Wu, Y., Li, J., Wang, X., Zeng, Z., Xu, J., Liu, Y., Feng, J., Chen, H., He, Y., Xia, R., 2023. TBtools-II: a "one for all, all for one" bioinformatics platform for biological big-data mining. *Mol. Plant.* <https://doi.org/10.1016/j.molp.2023.09.010>.
- Chen, J., Chang, S.X., Anyia, A.O., 2011. The physiology and stability of leaf carbon isotope discrimination as a measure of water-use efficiency in Barley on the Canadian prairies. *J. Agron. Crop Sci.* 191, 1–10.
- Ciftci-Yilmaz, S., Mittler, R., 2008. The zinc finger network of plants. *Cell. Mol. Life Sci.* 65 (7), 1150–1160. <https://doi.org/10.1007/s00018-007-7473-4>.
- Davletova, S., Schlauch, K., Coutu, J., Mittler, R., 2005. The zinc-finger protein Zat12 plays a central role in reactive oxygen and abiotic stress signaling in *Arabidopsis*. *Plant Physiol.* 139 (2), 847–856. <https://doi.org/10.1104/pp.105.068254>.
- Devin, S.R., Prudencio, A.S., Mahdavi, S.M.E., Rubio, M., Martínez-García, P.J., Martínez-Gómez, P., 2023. Orchard management and incorporation of biochemical and molecular strategies for improving drought tolerance in fruit tree crops. *Plants* 12 (4). <https://doi.org/10.3390/plants12040773>.
- Englbrecht, C.C., Schoof, H., Böhm, S., 2004. Conservation, diversification and expansion of C2H2 zinc finger proteins in the *Arabidopsis thaliana* genome. *BMC Genom.* 5 (1), 39. <https://doi.org/10.1186/1471-2164-5-39>.
- Farquhar, G.D., O'Leary, M.H., Berry, J.A., 1982. On the relationship between carbon isotope discrimination and the intercellular carbon dioxide concentration in leaves. *Aust. J. Plant Physiol.* 13 (2), 281–292.
- Geng, D., Chen, P., Shen, X., Zhang, Y., Li, X., Jiang, L., Xie, Y., Niu, C., Zhang, J., Huang, X., Ma, F., Guan, Q., 2018. MdMYB88 and MdMYB124 enhance drought tolerance by modulating root vessels and cell walls in apple. *Plant Physiol.* 178 (3), 1296–1309. <https://doi.org/10.1104/pp.18.00502>.
- González, E.M., 2023. Drought stress tolerance in plants. *Int. J. Mol. Sci.* 24 (7). <https://doi.org/10.3390/ijms24076562>.
- Han, G., Li, Y., Qiao, Z., Wang, C., Zhao, Y., Guo, J., Chen, M., Wang, B., 2021. Advances in the regulation of epidermal cell development by C2H2 zinc finger proteins in plants. *Front. Plant Sci.* 12, 754512. <https://doi.org/10.3389/fpls.2021.754512>.
- Han, G., Lu, C., Guo, J., Qiao, Z., Sui, N., Qiu, N., Wang, B., 2020. C2H2 zinc finger proteins: master regulators of abiotic stress responses in plants. *Front. Plant Sci.* 11, 115. <https://doi.org/10.3389/fpls.2020.00115>.
- Kielbowicz-Matuk, A., 2012. Involvement of plant C(2)H(2)-type zinc finger transcription factors in stress responses. *Plant Sci.* 185–186, 78–85. <https://doi.org/10.1016/j.plantsci.2011.11.015>.
- Kodaira, K.S., Qin, F., Tran, L.S., Maruyama, K., Kidokoro, S., Fujita, Y., Shinozaki, K., Yamaguchi-Shinozaki, K., 2011. *Arabidopsis* Cys2/His2 zinc-finger proteins AZF1 and AZF2 negatively regulate abscisic acid-repressive and auxin-inducible genes under abiotic stress conditions. *Plant Physiol.* 157 (2), 742–756. <https://doi.org/10.1104/pp.111.182683>.
- Lee, D.K., Jung, H., Jiang, G., Jeong, J.S., Kim, Y.S., Ha, S.H., Do Choi, Y., Kim, J.K., 2016. Overexpression of the OsERF71 transcription factor alters rice root structure and drought resistance. *Plant Physiol.* 172 (1), 575–588. <https://doi.org/10.1104/pp.16.00379>.
- Li, H., Yue, M., Jiang, L., Liu, Y., Zhang, N., Liu, X., Ye, Y., Lin, X., Zhang, Y., Lin, Y., Li, M., Wang, Y., Zhang, Y., Luo, Y., Wang, X., Chen, Q., Tang, H., 2022. Genome-Wide identification of strawberry C2H2-ZFP C1-2i subclass and the potential function of FaZAT10 in abiotic stress. *Int. J. Mol. Sci.* 23 (21). <https://doi.org/10.3390/ijms232113079>.
- Li, X., Ma, Z., Song, Y., Shen, W., Yue, Q., Khan, A., Tahir, M.M., Wang, X., Malnoy, M., Ma, F., Bus, V., Zhou, S., Guan, Q., 2023. Insights into the molecular mechanisms underlying responses of apple trees to abiotic stresses. *Hortic. Res.* 10 (8), uhad144. <https://doi.org/10.1093/hr/uhad144>.

- Liu, Y., Khan, A.R., Gan, Y., 2022. C2H2 zinc finger proteins response to abiotic stress in plants. *Int. J. Mol. Sci.* 23 (5) <https://doi.org/10.3390/ijms23052730>.
- Lux, A., Rost, T.L., 2012. Plant root research: the past, the present and the future. *Ann. Bot.* 110 (2), 201–204. <https://doi.org/10.1093/aob/mcs156>.
- Manna, M., Thakur, T., Chirom, O., Mandlik, R., Deshmukh, R., Salvi, P., 2021. Transcription factors as key molecular target to strengthen the drought stress tolerance in plants. *Physiol. Plantarum* 172 (2), 847–868. <https://doi.org/10.1111/ppl.13268>.
- Medrano, Hipolito, Hernandez-Montes, Esther, Tortosa, Ignacio, Brahama, Mohamed, Bchir, Amani, 2016. Carbon isotope discrimination ($\delta^{13}C$) as an indicator of vine water status and water use efficiency (WUE): looking for the most representative sample and sampling time. *Agric. Water Manag.* 167, 11–20.
- Niu, C., Li, H., Jiang, L., Yan, M., Li, C., Geng, D., Xie, Y., Yan, Y., Shen, X., Chen, P., Dong, J., Ma, F., Guan, Q., 2019. Genome-wide identification of drought-responsive microRNAs in two sets of *Malus* from interspecific hybrid progenies. *Hortic. Res.* 6, 75. <https://doi.org/10.1038/s41438-019-0157-z>.
- Rowland, L., Ramírez-Valiente, J.A., Hartley, I.P., Mencuccini, M., 2023. How woody plants adjust above- and below-ground traits in response to sustained drought. *New Phytol.* 239 (4), 1173–1189. <https://doi.org/10.1111/nph.19000>.
- Seki, M., Umezawa, T., Urano, K., Shinozaki, K., 2007. Regulatory metabolic networks in drought stress responses. *Curr. Opin. Plant Biol.* 10 (3), 296–302. <https://doi.org/10.1016/j.pbi.2007.04.014>.
- Singh, K., Foley, R.C., Oñate-Sánchez, L., 2002. Transcription factors in plant defense and stress responses. *Curr. Opin. Plant Biol.* 5 (5), 430–436. [https://doi.org/10.1016/s1369-5266\(02\)00289-3](https://doi.org/10.1016/s1369-5266(02)00289-3).
- Tang, W., Luo, C., 2018. Overexpression of zinc finger transcription factor ZAT6 enhances salt tolerance. *Open Life Sci.* 13, 431–445. <https://doi.org/10.1515/biol-2018-0052>.
- Tamura, K., Stecher, G., Kumar, S., 2021. MEGA11: molecular evolutionary genetics analysis version 11. *Mol. Biol. Evol.* 38, 3022–3027.
- Wan, S., Liang, B., Yang, L., Hu, W., Kuang, L., Song, J., Xie, J., Huang, Y., Liu, D., Liu, Y., 2023. The MADS-box family gene *PtrANR1* encodes a transcription activator promoting root growth and enhancing plant tolerance to drought stress. *Plant Cell Rep.* 43 (1), 16. <https://doi.org/10.1007/s00299-023-03121-7>.
- Wang, H., Zhao, S., Mao, K., Dong, Q., Liang, B., Li, C., Wei, Z., Li, M., Ma, F., 2018. Mapping QTLs for water-use efficiency reveals the potential candidate genes involved in regulating the trait in apple under drought stress. *BMC Plant Biol.* 18 (1), 136. <https://doi.org/10.1186/s12870-018-1308-3>.
- Wang, K., Ding, Y., Cai, C., Chen, Z., Zhu, C., 2019. The role of C2H2 zinc finger proteins in plant responses to abiotic stresses. *Physiol. Plantarum* 165 (4), 690–700. <https://doi.org/10.1111/ppl.12728>.
- Xie, Y., Bao, C., Chen, P., Cao, F., Liu, X., Geng, D., Li, Z., Li, X., Hou, N., Zhi, F., Niu, C., Zhou, S., Zhan, X., Ma, F., Guan, Q., 2021. Absciscic acid homeostasis is mediated by feedback regulation of MdMYB88 and MdMYB124. *J. Exp. Bot.* 72 (2), 592–607. <https://doi.org/10.1093/jxb/eraa449>.
- Xie, Y., Chen, P., Yan, Y., Bao, C., Li, X., Wang, L., Shen, X., Li, H., Liu, X., Niu, C., Zhu, C., Fang, N., Shao, Y., Zhao, T., Yu, J., Zhu, J., Xu, L., van Nocker, S., Ma, F., Guan, Q., 2018. An atypical R2R3 MYB transcription factor increases cold hardiness by CBF-dependent and CBF-independent pathways in apple. *New Phytol.* 218 (1), 201–218. <https://doi.org/10.1111/nph.14952>.
- Yang, K., Li, C.Y., An, J.P., Wang, D.R., Wang, X., Wang, C.K., You, C.X., 2021. The C2H2-type zinc finger transcription factor MdZAT10 negatively regulates drought tolerance in apple. *Plant Physiol. Biochem.* 167, 390–399. <https://doi.org/10.1016/j.plaphy.2021.08.014>.
- Zhang, D., He, J., Cheng, P., Zhang, Y., Khan, A., Wang, S., Li, Z., Zhao, S., Zhan, X., Ma, F., Li, X., Guan, Q., 2023. 4-methylumbelliferone (4-MU) enhances drought tolerance of apple by regulating rhizosphere microbial diversity and root architecture. *Hortic. Res.* 10 (6), uhad099 <https://doi.org/10.1093/hr/uhad099>.
- Zhao, P., Hou, S., Guo, X., Jia, J., Yang, W., Liu, Z., Chen, S., Li, X., Qi, D., Liu, G., Cheng, L., 2019. A MYB-related transcription factor from sheepgrass, LcMYB2, promotes seed germination and root growth under drought stress. *BMC Plant Biol.* 19 (1), 564. <https://doi.org/10.1186/s12870-019-2159-2>.